

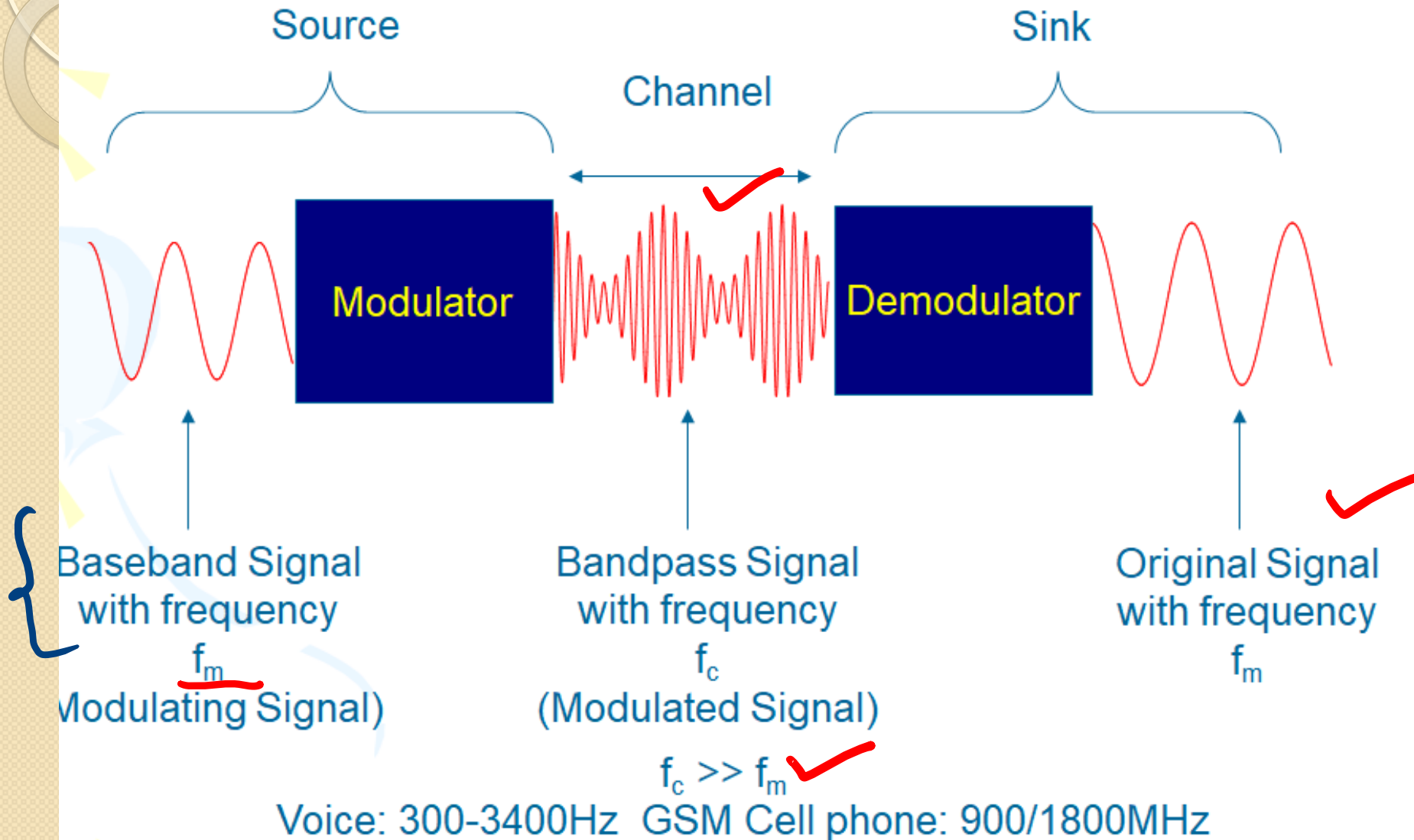


Unit 1.2

Amplitude Modulation

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AM Modulation and Demodulation

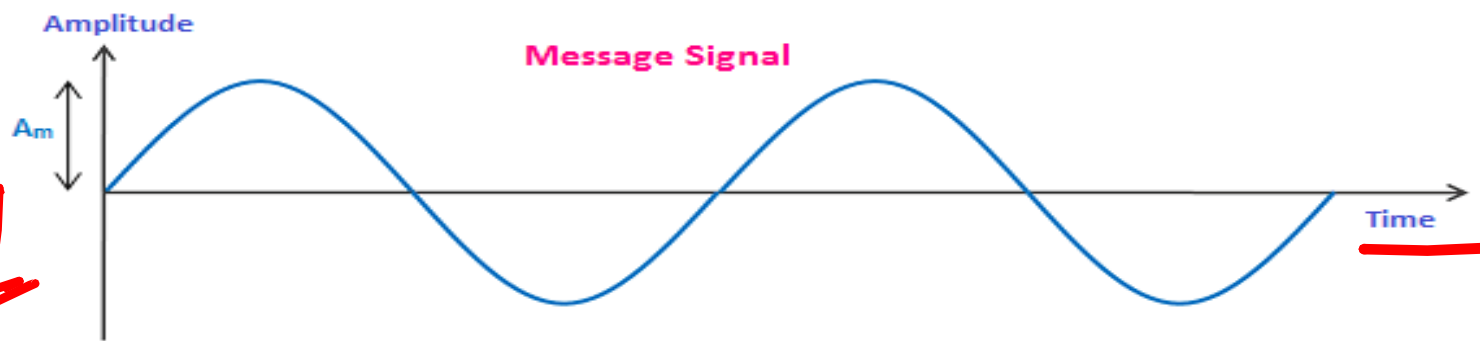


Definition of Modulation

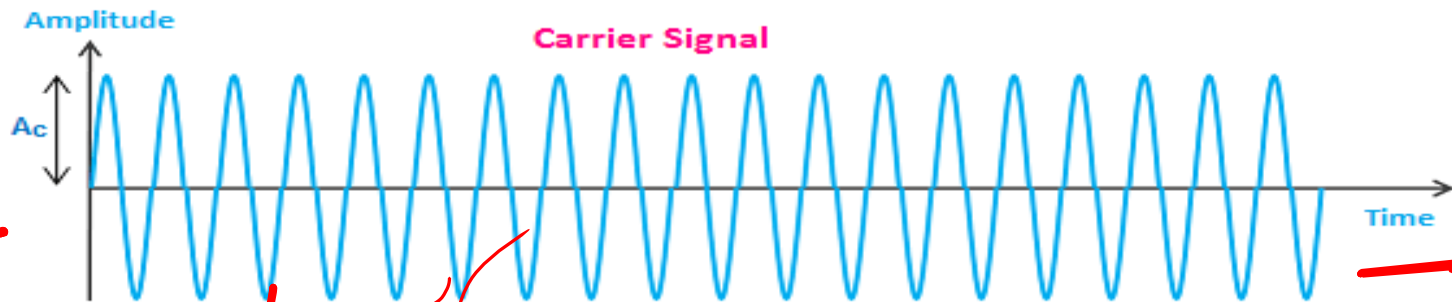
- A Modulation is the process by which **some characteristics of a carrier wave** is varied in accordance with the instantaneous value of a **modulating signal or message signal**.
- The carrier wave is changed in accordance with the **intensity (i.e amplitude) of the message signal**.
- The resultant wave is called **modulated wave** or radio wave.

①

$V_m = 3V$

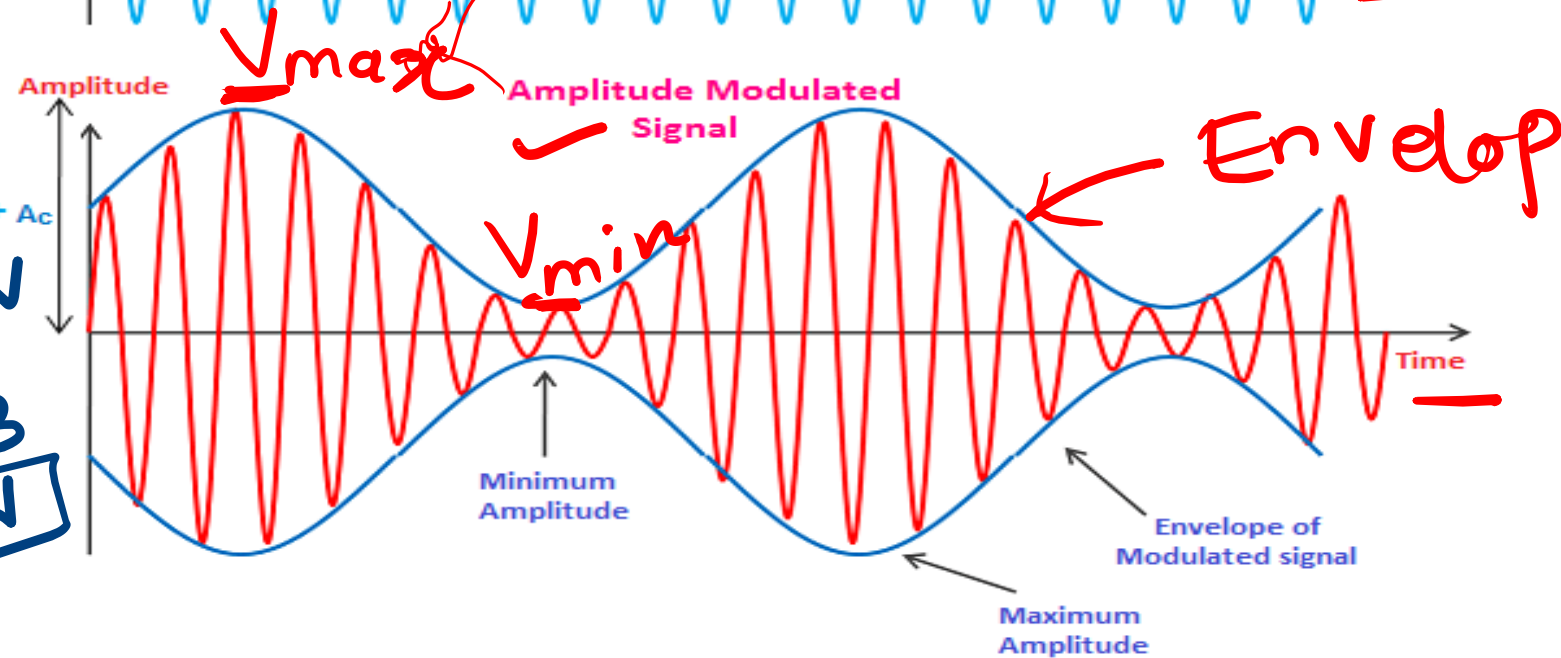


$V_c = 6V$



$V_{max} = A_m + A_c$
 $6 + 3 = 9V$

$V_{min} = 6 - 3 = 3V$



Amplitude Modulation Basics

Process of changing **amplitude** of carrier signals in accordance with instantaneous values of **modulating signal**(information), keeping **frequency** and **phase** of carrier signal **constant**

V_c
 f_m

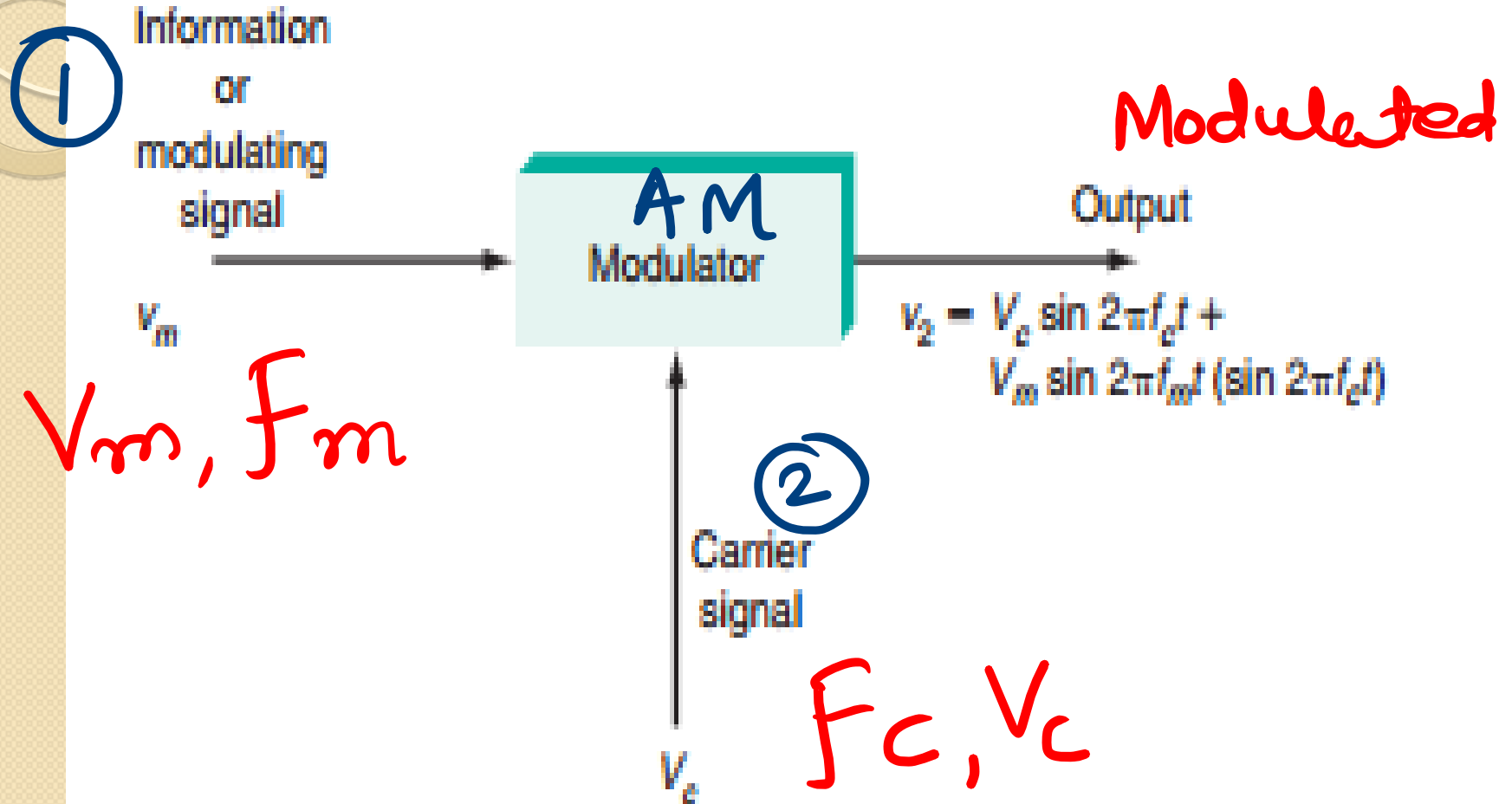
Amplitude Modulation Basics

- AM is relatively inexpensive, low quality form of modulation.
- Used for commercial broadcasting of both audio and video signals.
- Two way mobile communications such as citizens band (CB) radio.

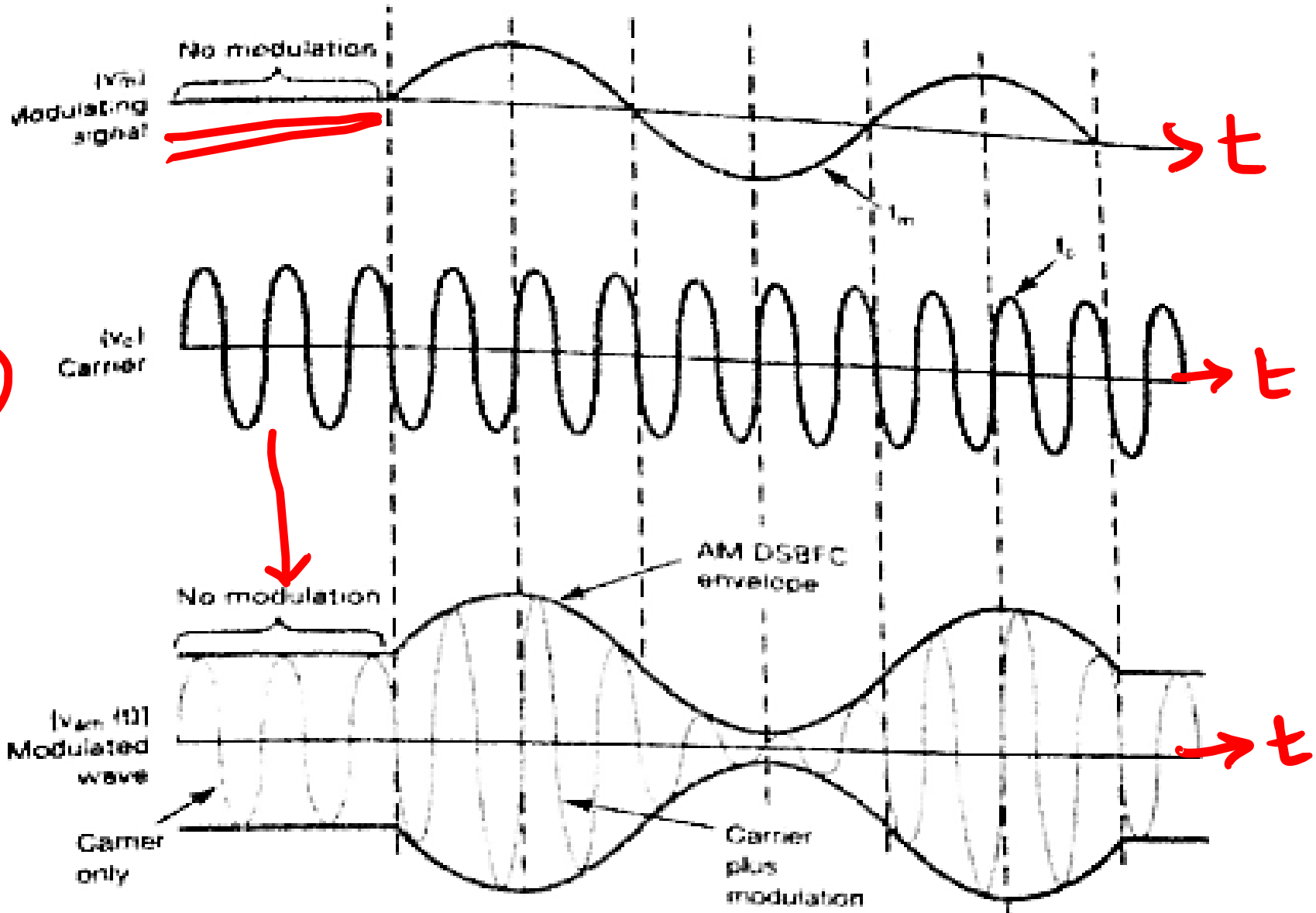
Amplitude Modulation basics

- Modulators are non linear device
- One input is a single high frequency carrier of constant amplitude
- And second input is low frequency signal.
- Voice communication utilizes range of information frequencies between 300Hz to 3000Hz.

Amplitude Modulation basics



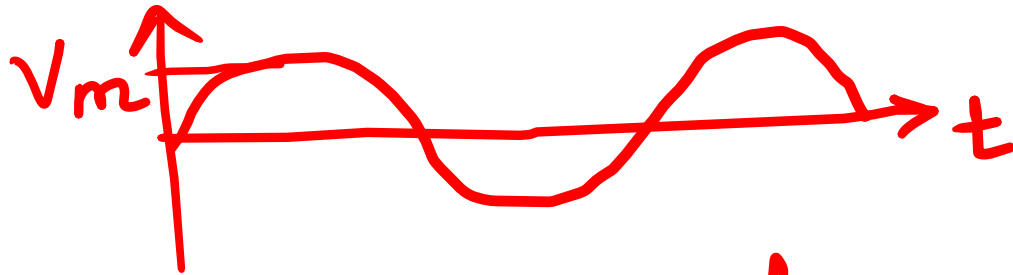
$$v_{AM}(t) = [V_c + V_m \sin(2\pi f_m t)][\sin(2\pi f_c t)]$$



Derivation of AM

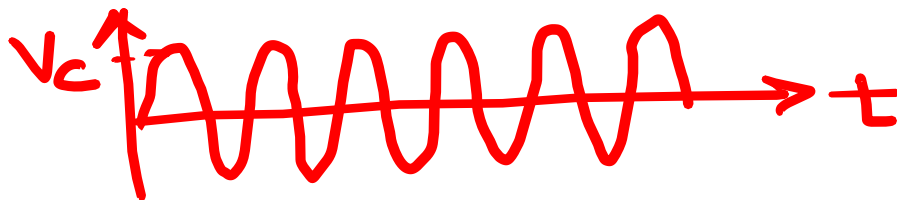
→ Let modulating signal

$$V_m = V_m \sin \omega_m t \quad \text{--- (1)}$$



→ Let carrier signal

$$V_c = V_c \sin \omega_c t \quad \text{--- (2)}$$



~~$2\pi f_m$~~ $2\pi f_m$

A green arrow points from the $2\pi f_m$ term in the equation $V_m = V_m \sin \omega_m t$ to the $2\pi f_m$ term in this block. The $2\pi f_m$ term is circled in green, and the f_m is crossed out with a green 'X'.

Derivation of AM

Let amplitude of Am wave is

$$V_{AM} = V_c + V_m \text{ ---- (Definition)}$$

$$V_{AM} = V_c + V_m \sin \omega_m t$$

$$V_{AM} = V_c \left[1 + \frac{V_m}{V_c} \sin \omega_m t \right]$$

25%
↑

$$\text{Modulation Index } (m) = \frac{V_m}{V_c} \times 100 \text{ Percentage}$$

Derivation of AM

$$V_{AM} = V_c [1 + m \sin \omega_m t] \quad \text{--- (3)}$$

$$V_{AM}(t) = V_{AM} \sin \omega_c t$$

$\xrightarrow{V_c} V_c + V_m$

$$V_{AM}(t) = V_c [1 + m \sin \omega_m t] \sin \omega_c t$$

$$V_{AM}(t) = V_c \sin \omega_c t + m V_c \underbrace{\sin \omega_c t \cdot \sin \omega_m t}_{\sin A \cdot \sin B}$$

$$\boxed{\sin A \cdot \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]}$$

Derivation of AM

$$V_{Am}(t) = \underbrace{V_c \sin \omega_c t}_{\text{Carrier Signal}} + \underbrace{\frac{mV_c}{2} \cos(\omega_c - \omega_m)t}_{\text{Lower Sideband (LSB)}}$$

Lower Sideband (LSB)

$$- \frac{mV_c}{2} \cos(\omega_c + \omega_m)t$$

Upper Sideband (USB)

Derivation of AM

Frequency Domain representation of AM

Bandwidth

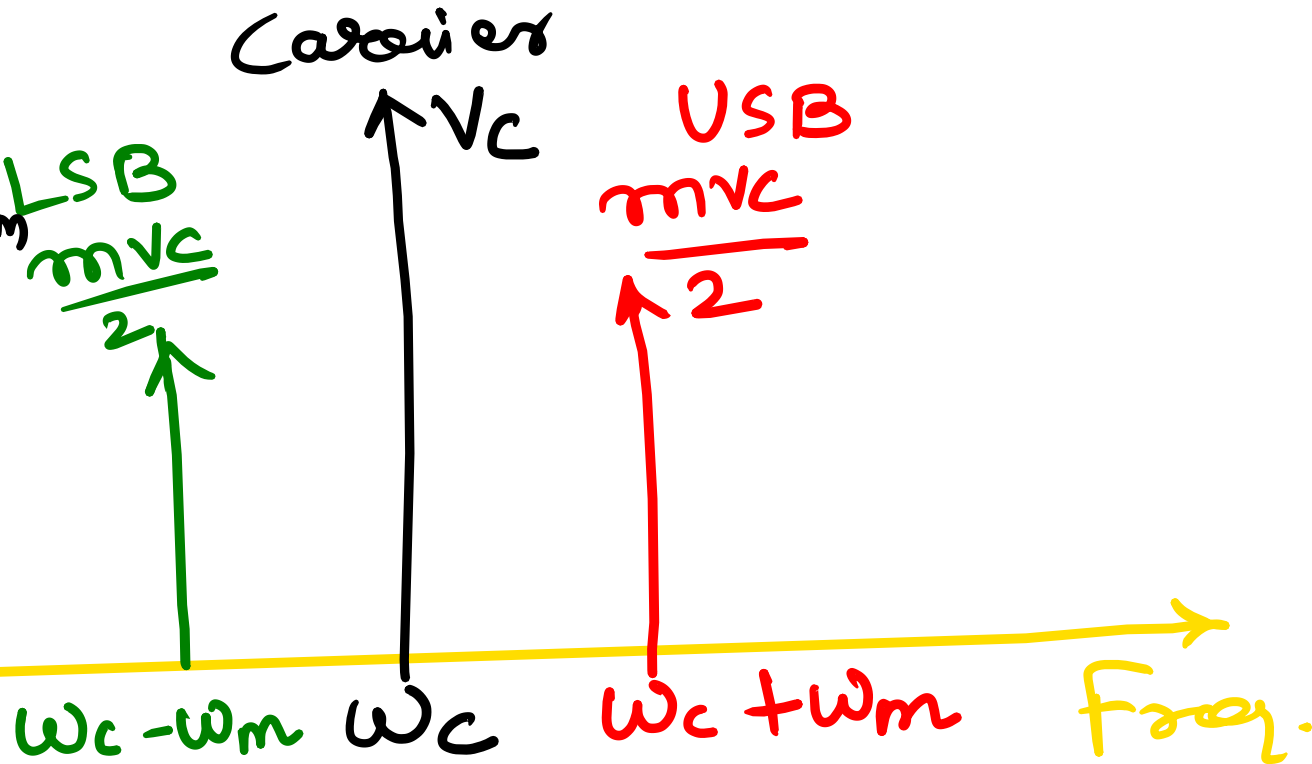
$$= \text{USB} - \text{LSB}$$

$$= \omega_c + \omega_m - \omega_c + \omega_m$$

$$B.W = 2\omega_m$$

or

$$B.W = 2f_m$$



Modulation Index

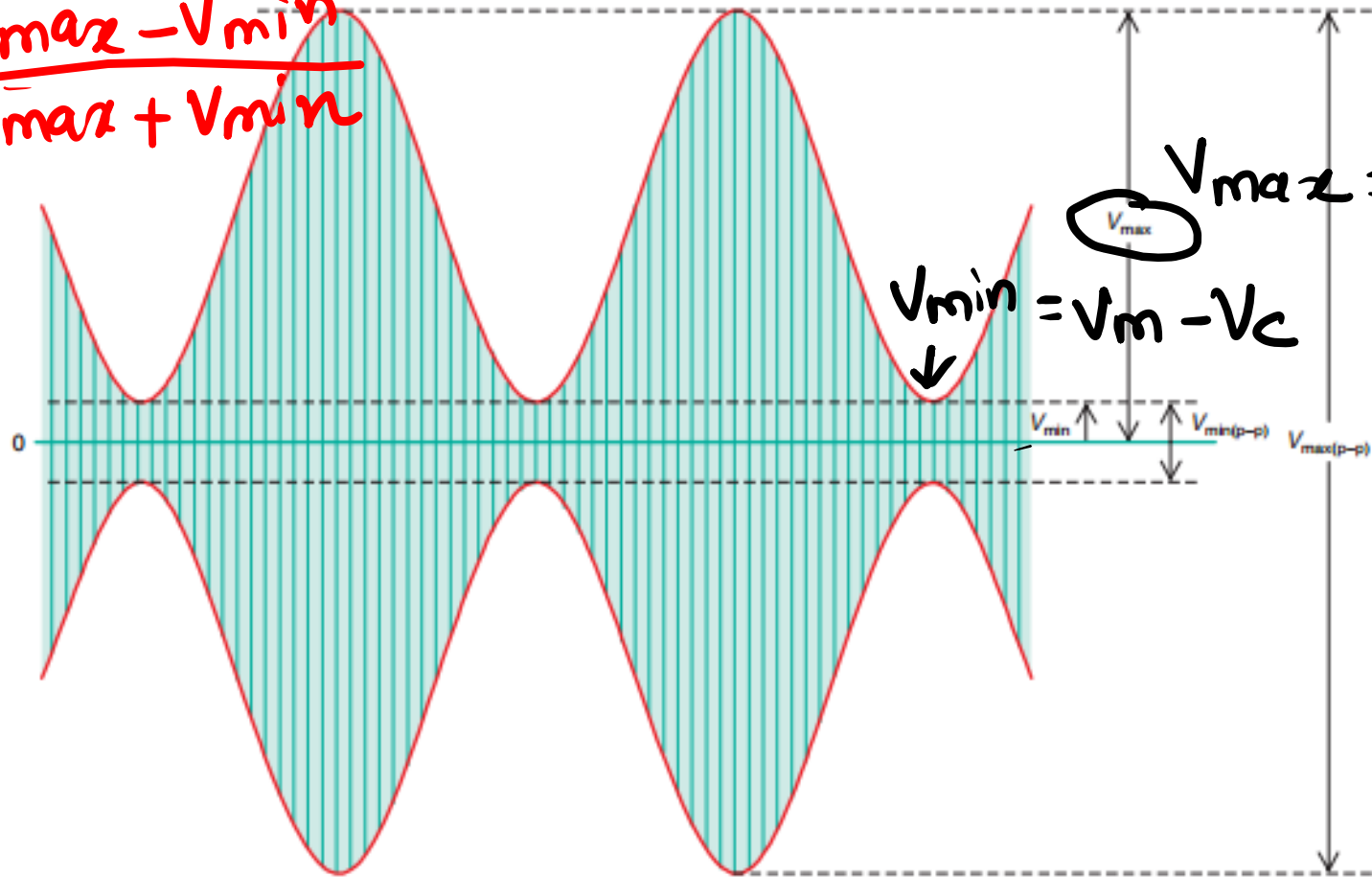
- It is Ratio of modulating signal voltage (V_m) to carrier signal voltage (V_c).

$$m = \frac{V_m}{V_c}$$

- Value of m will be between 0 and 1
- $m < 1$ is called under modulation
- $m = 1$ is called 100% modulation
- $m > 1$ is called over modulation

Modulation Index

$$m = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$$



$$V_{\max} = V_m + V_c$$
$$V_{\min} = V_m - V_c$$

Modulation Index

$$V_m = \frac{V_{\max} - V_{\min}}{2}$$

$$V_c = \frac{V_{\max} + V_{\min}}{2}$$

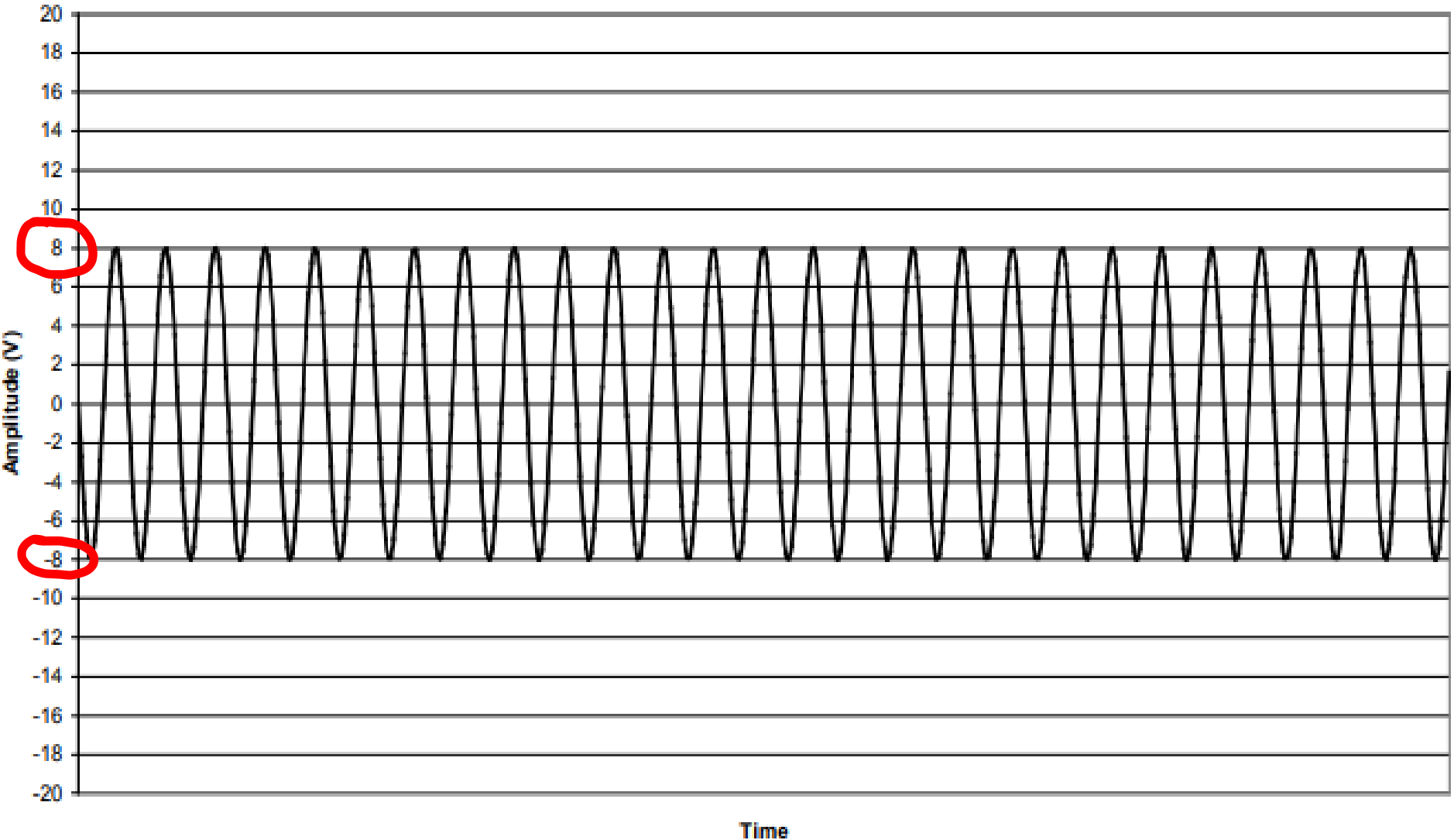
$$m = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$$



Depth of Modulation ($V_c=8V$)

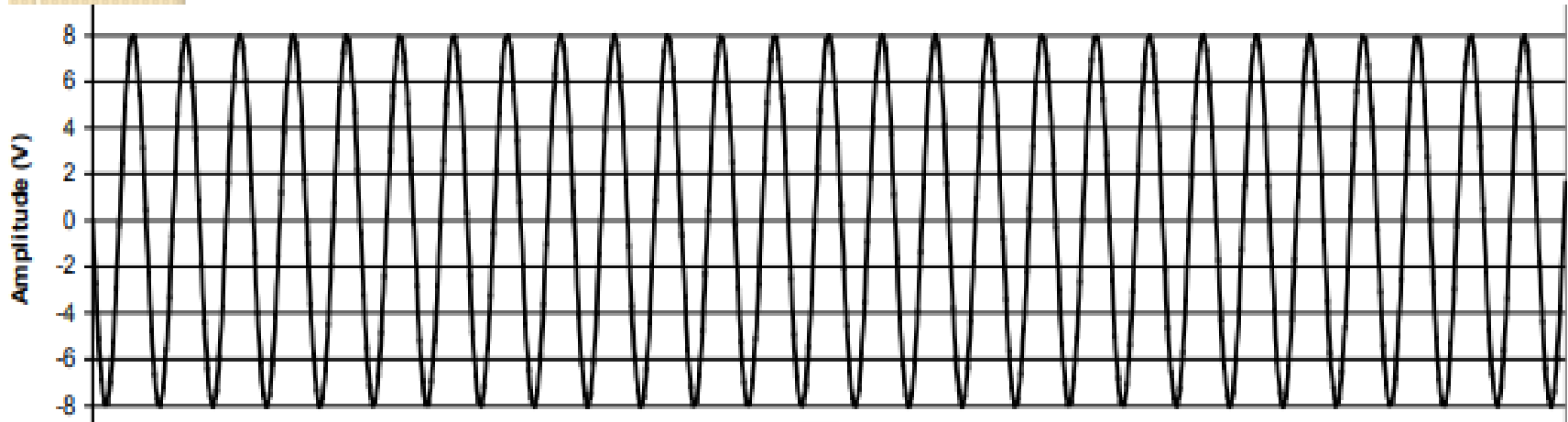
Peak

Carrier Wave

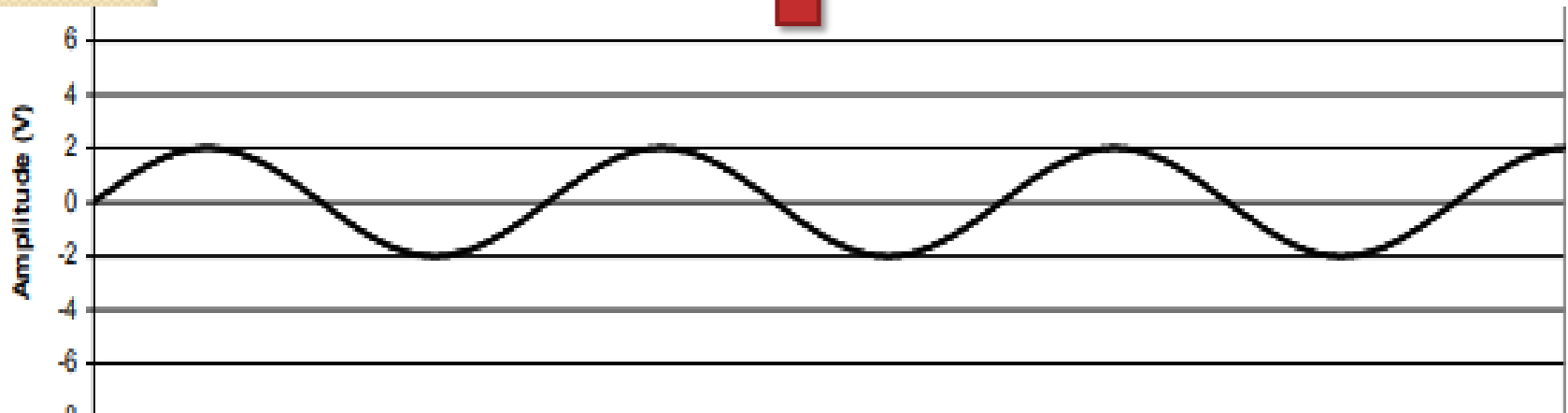


Case I: Add information signal of 2V to carrier

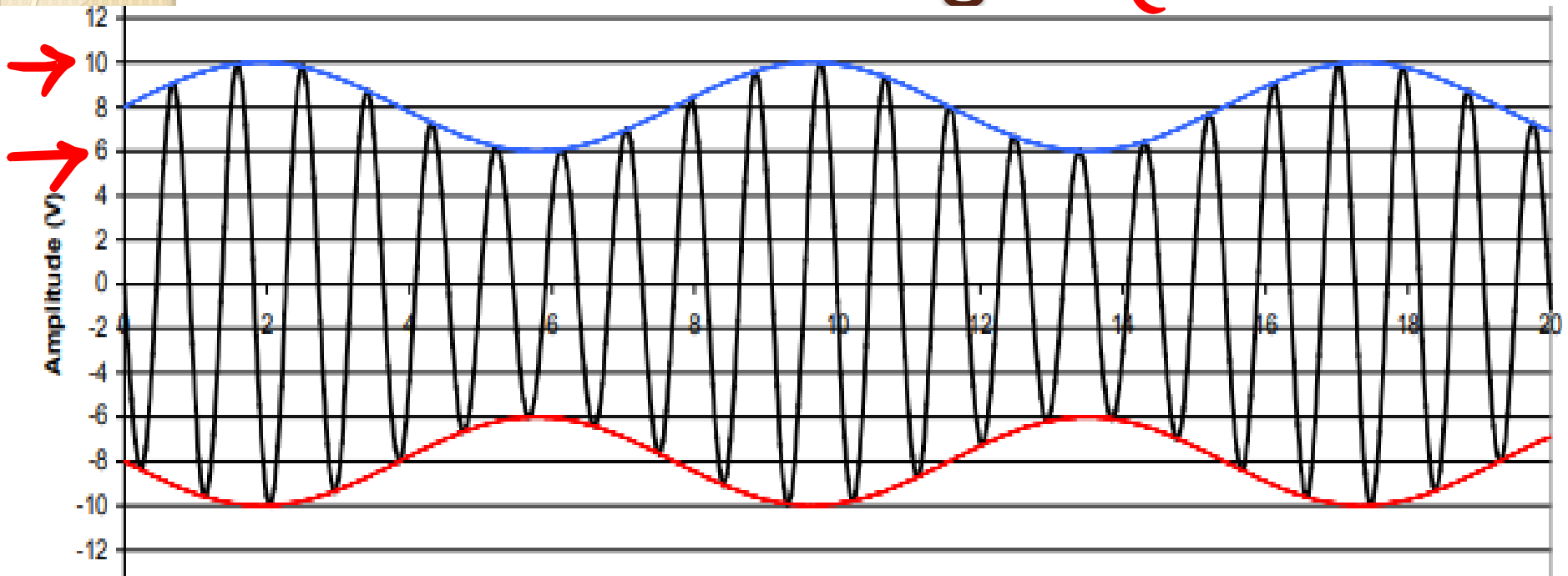
$V_c = 8V$



$V_m = 2V$



Resultant Signal (Modulated)



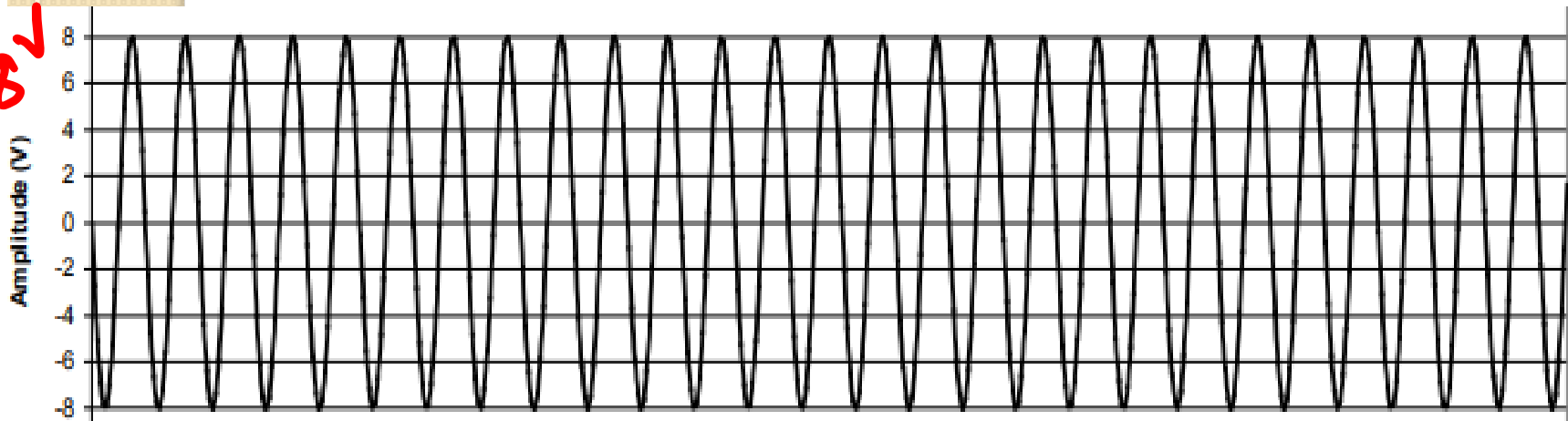
$$V_{max} = V_c + V_m = 8 + 2 = 10 \text{ V}$$

$$V_{min} = V_c - V_m = 8 - 2 = 6 \text{ V}$$

$$m = \frac{V_{max} - V_{min}}{V_{max} + V_{min}} = \frac{4}{16} \times 100\% = 25\%$$

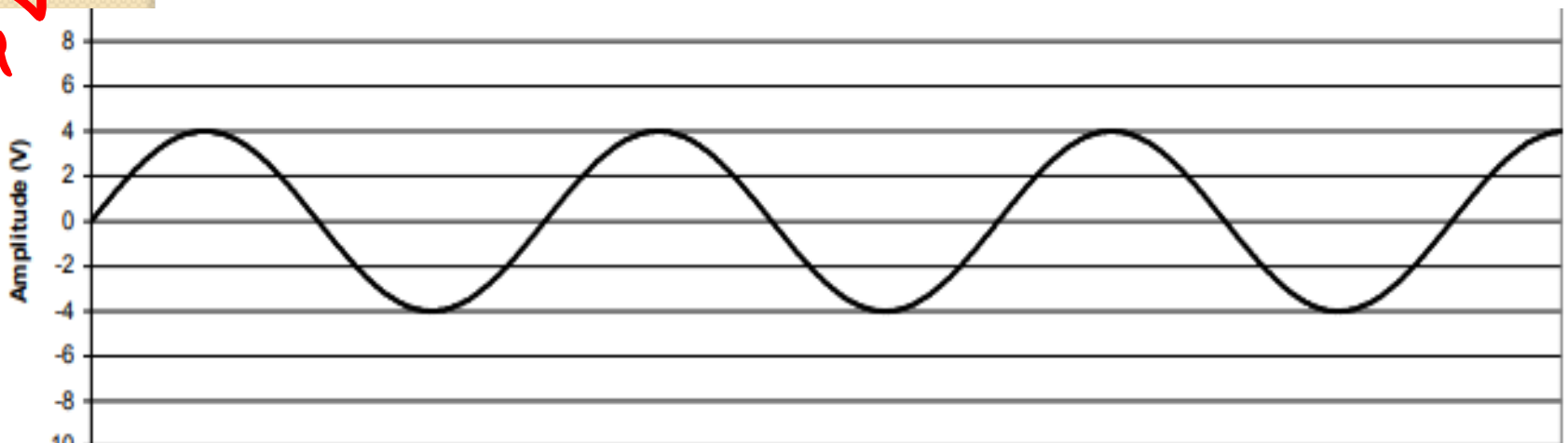
Case 2: Add information signal of 4V to carrier

$V_c = 8V$

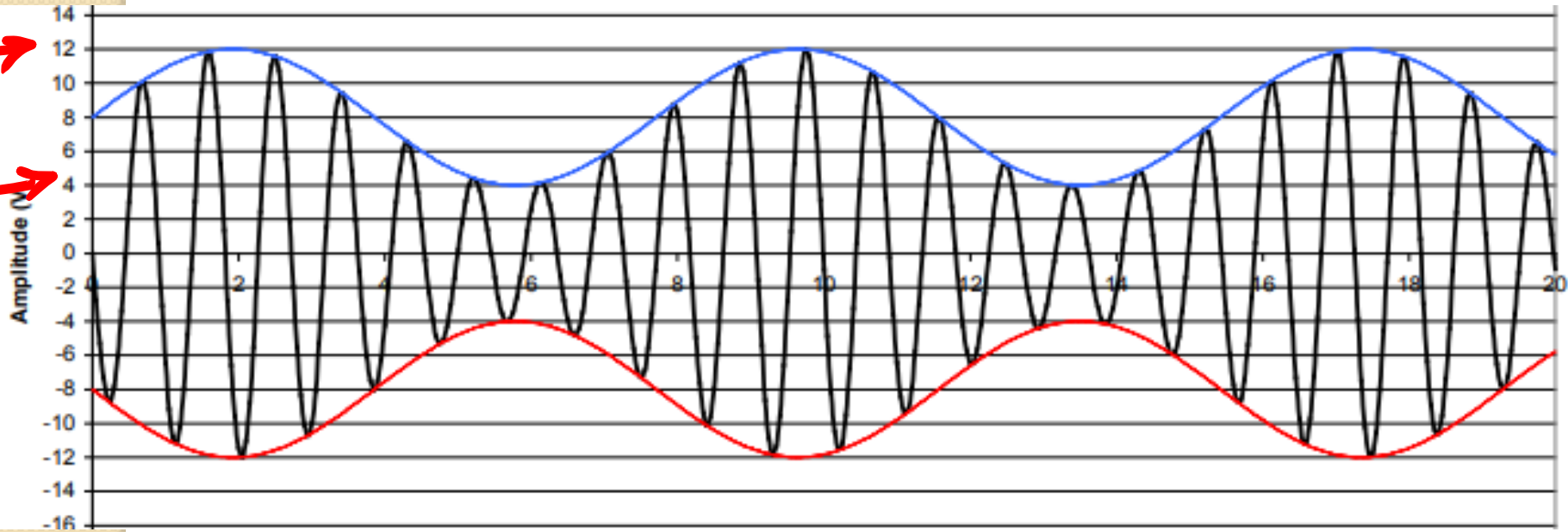


+

$V_m = 4V$



Resultant Signal



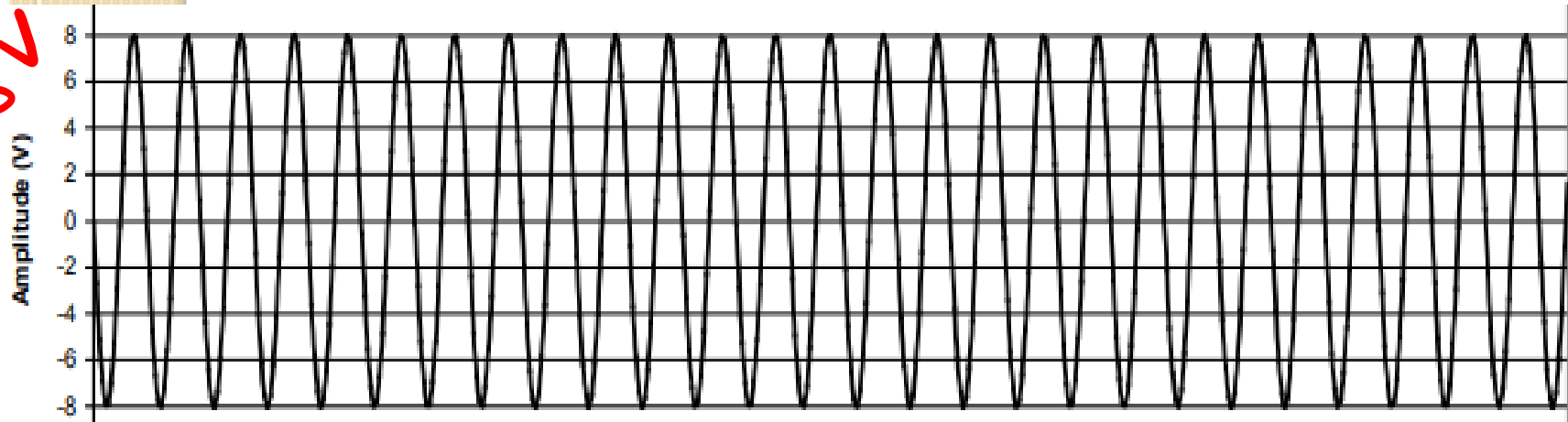
$$V_{\max} = V_c + V_m = 8 + 4 = 12 \text{ V}$$

$$V_{\min} = V_c - V_m = 8 - 4 = 4 \text{ V}$$

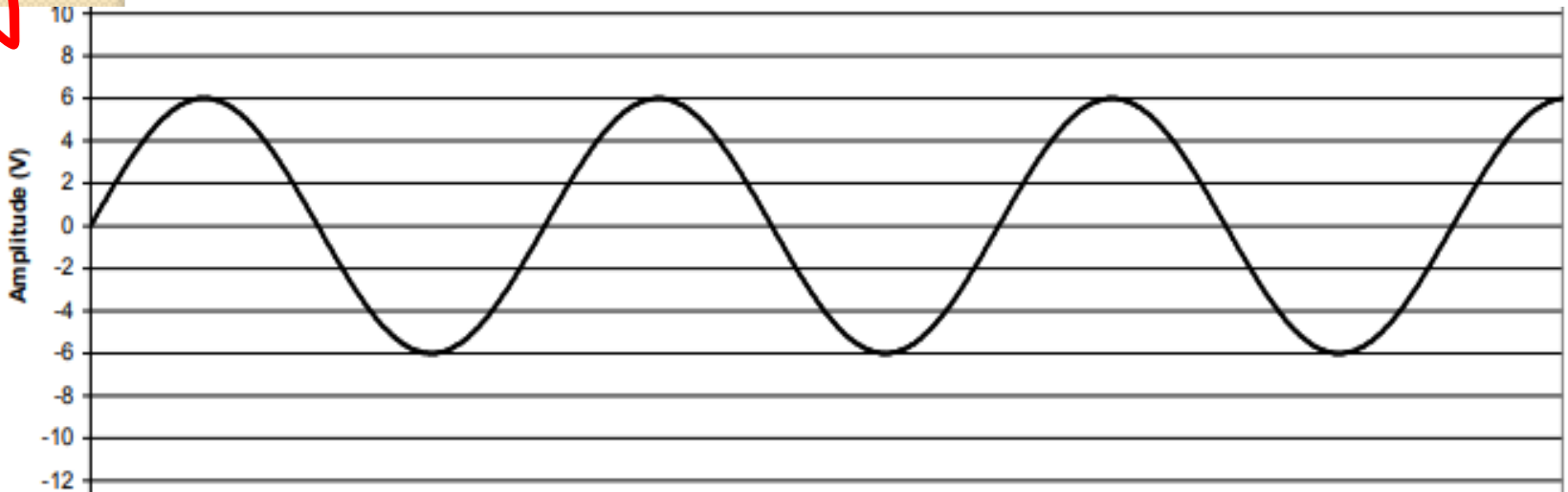
$$m = \frac{8}{16} \times 100\% = 50\%$$

Case 3: Add information signal of 6V to carrier

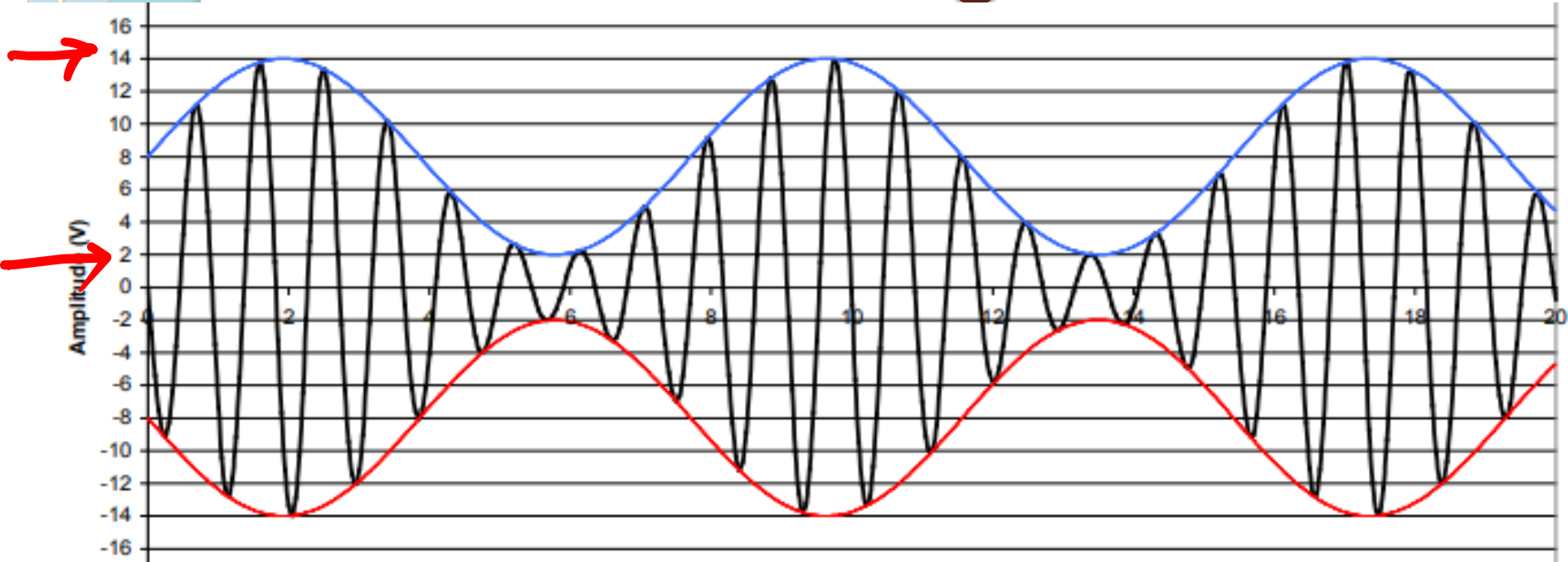
$V_c = 8V$



$V_m = 6V$



Resultant Signal



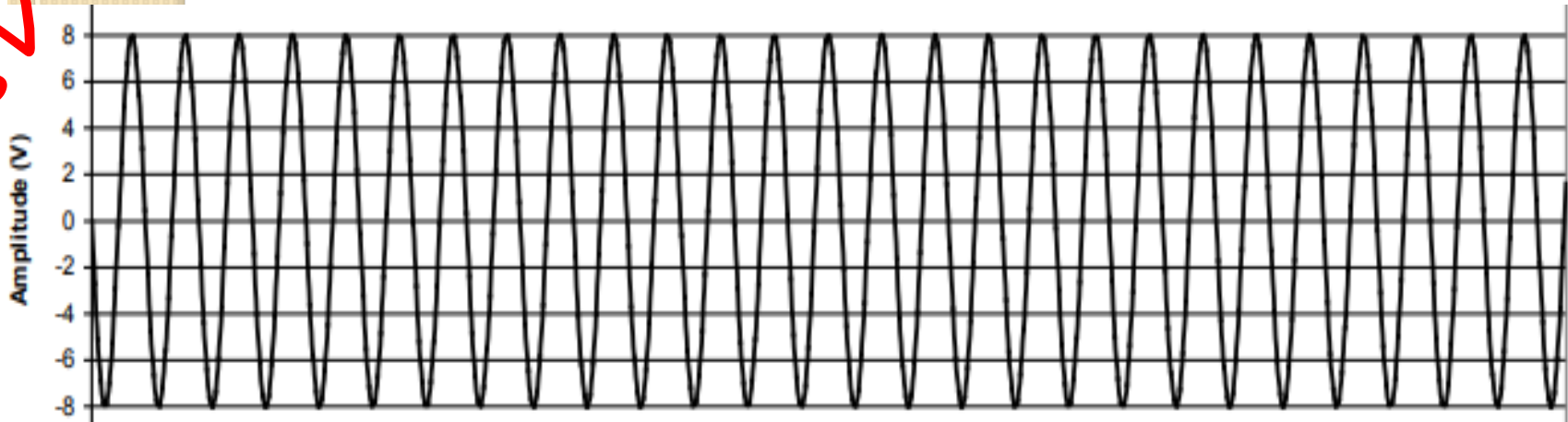
$$V_{\max} = V_c + V_m = 8 + 6 = 14 \text{ V}$$

$$V_{\min} = V_c - V_m = 8 - 6 = 2 \text{ V}$$

$$m = \frac{12}{16} \times 100\% = 75\%$$

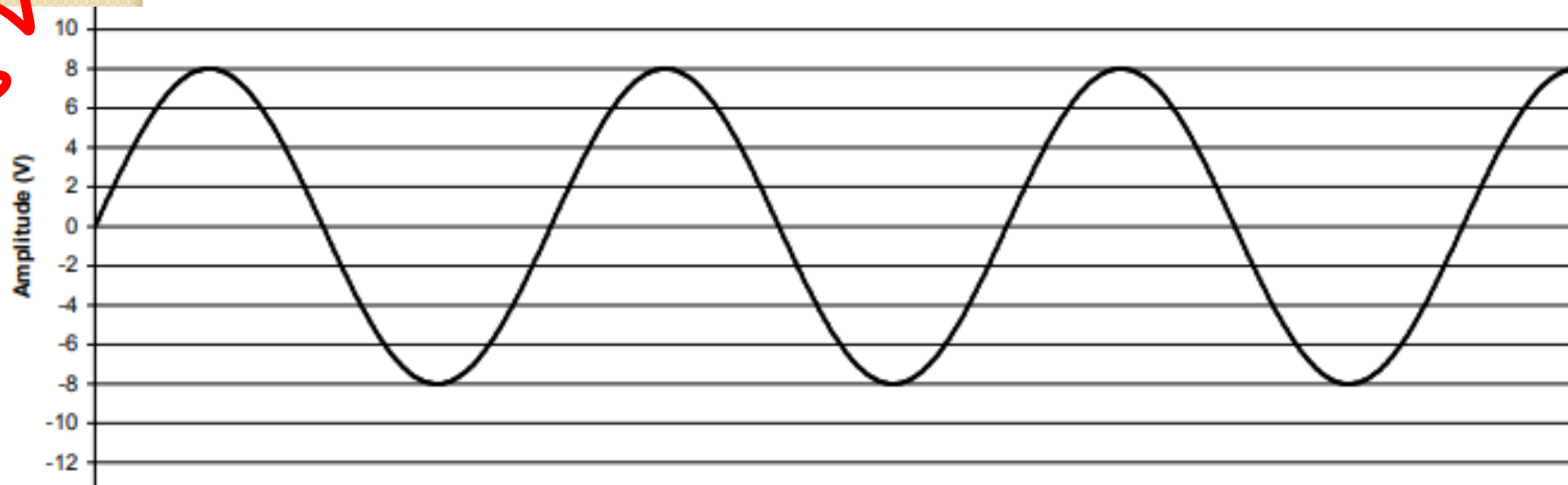
Case 4: Add information signal of 8V to carrier

$V_c = 8V$

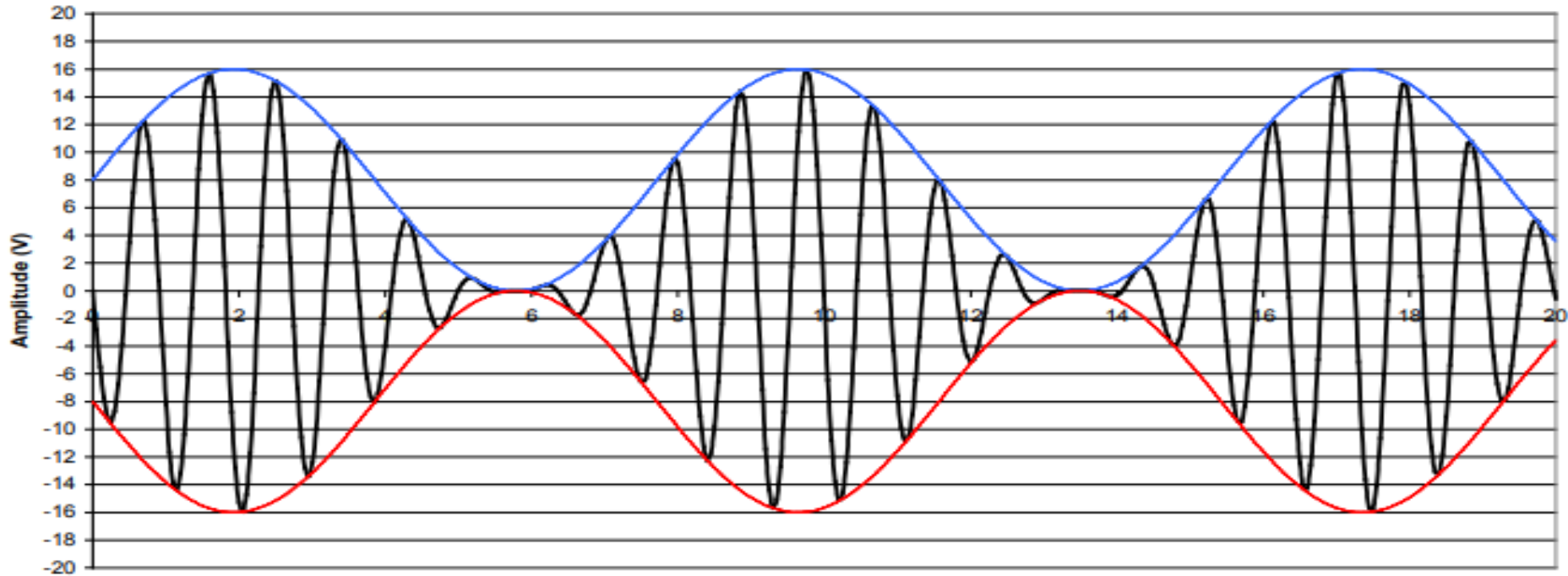


+

$V_m = 8V$



Resultant Signal



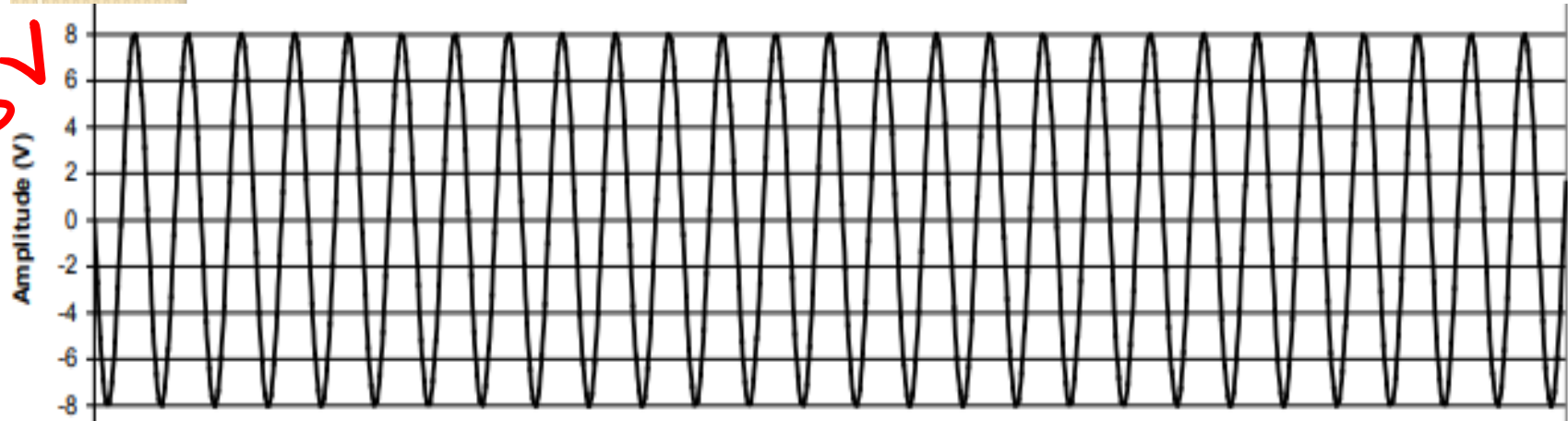
$$V_{\max} = V_c + V_m = 8 + 8 = 16 \text{ V}$$

$$V_{\min} = V_c - V_m = 8 - 8 = 0 \text{ V}$$

$$m = \frac{16}{16} \times 100\% = 100\%$$

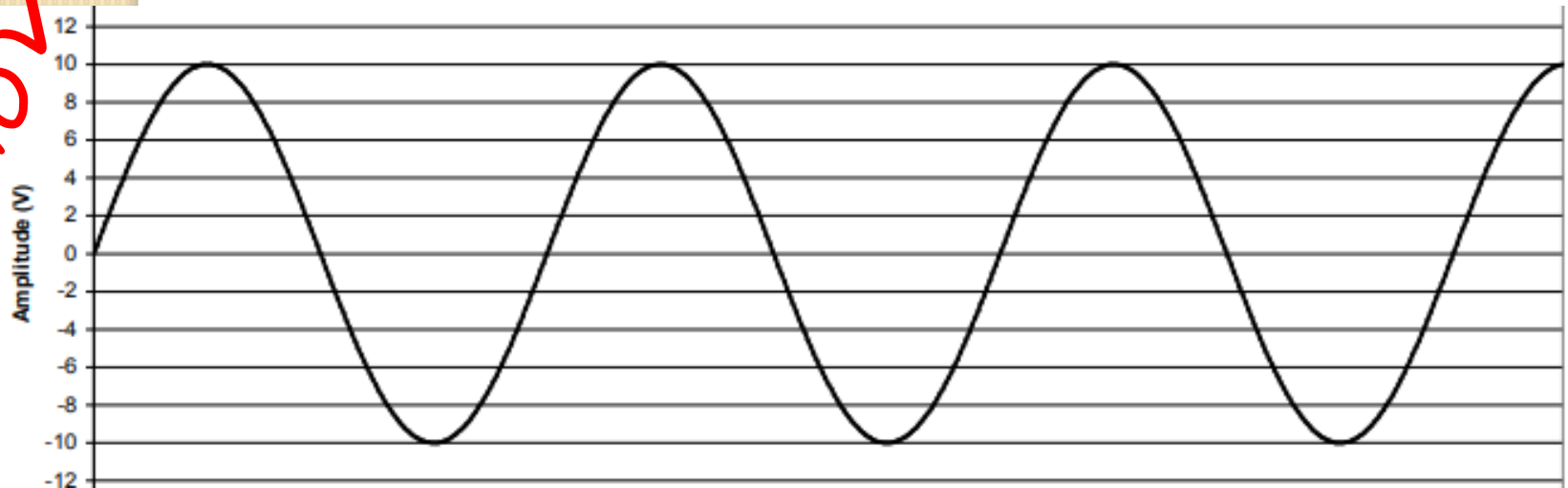
Case 5: Add information signal of 10V to carrier

$V_c = 8V$

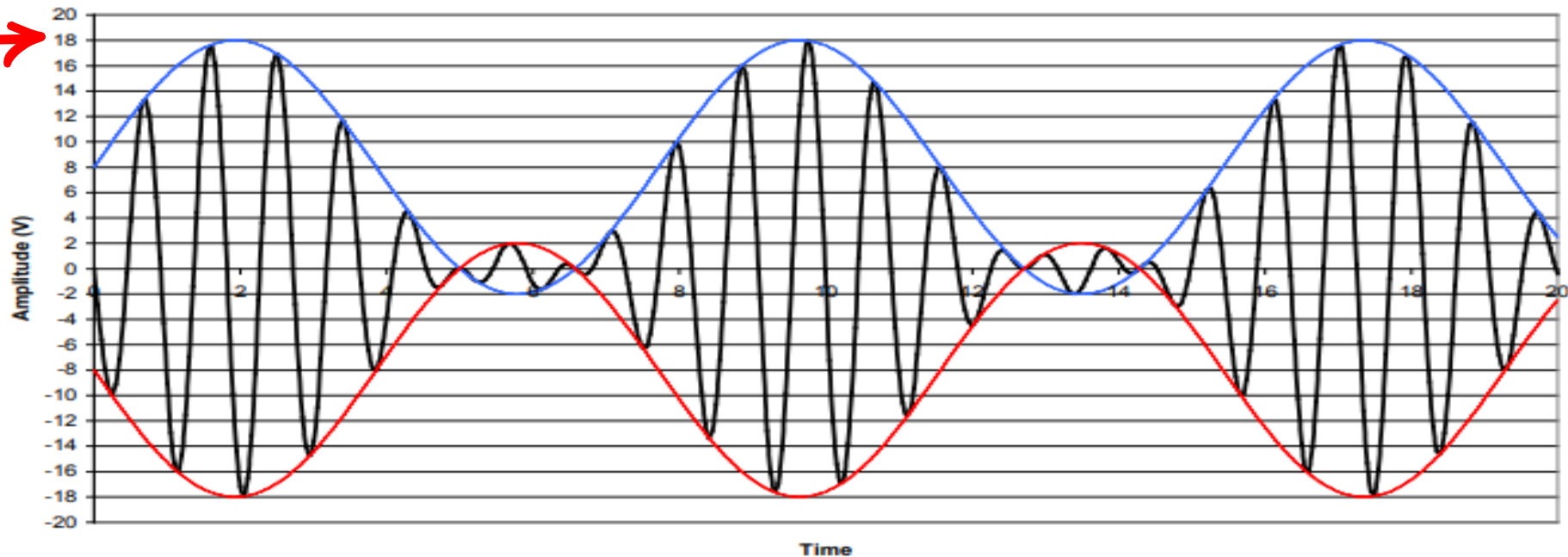


+

$V_m = 10V$



Resultant Signal



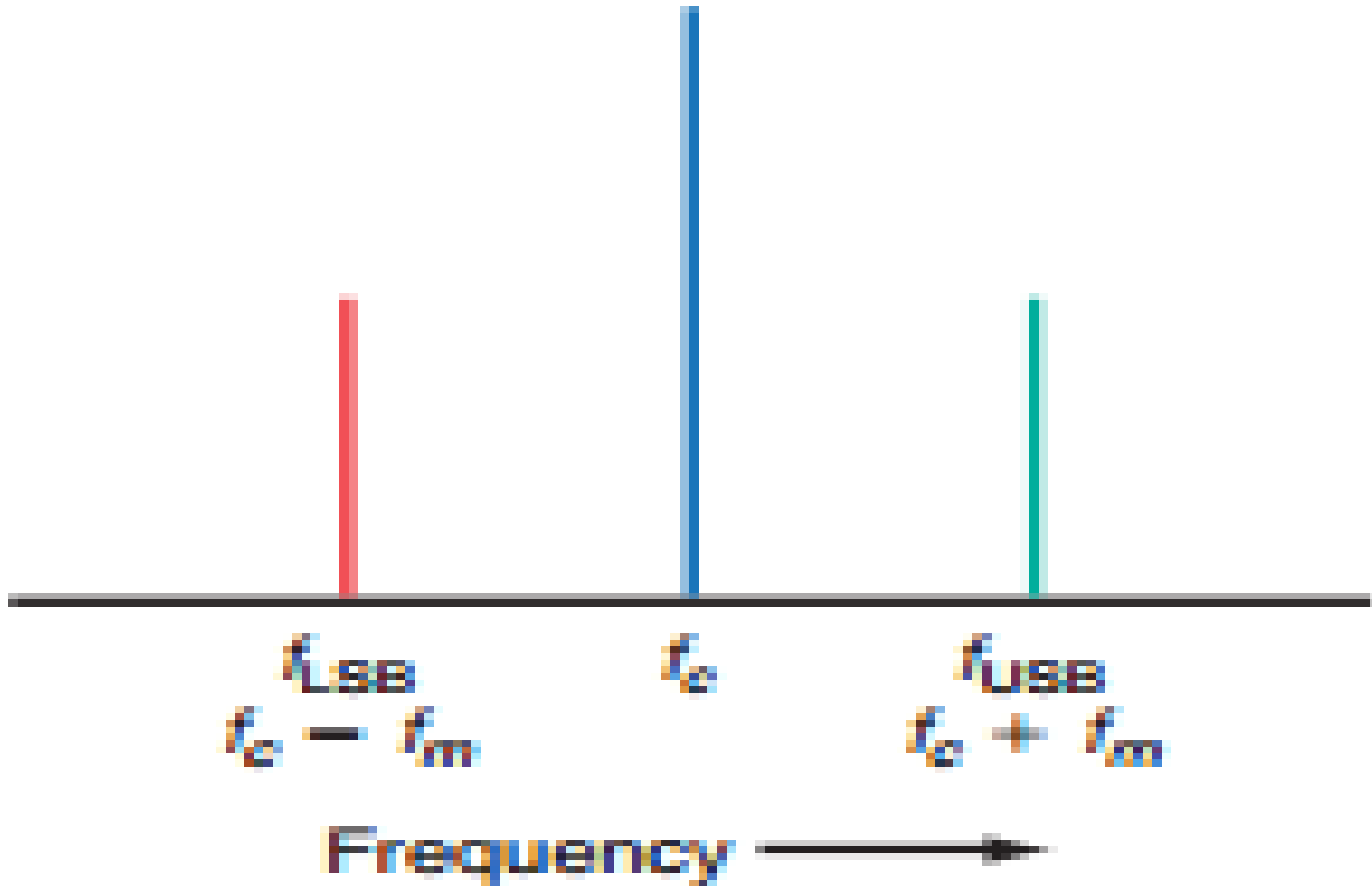
Over Modulation

$$V_{\max} = V_c + V_m = 8 + 10 = 18 \text{ V}$$

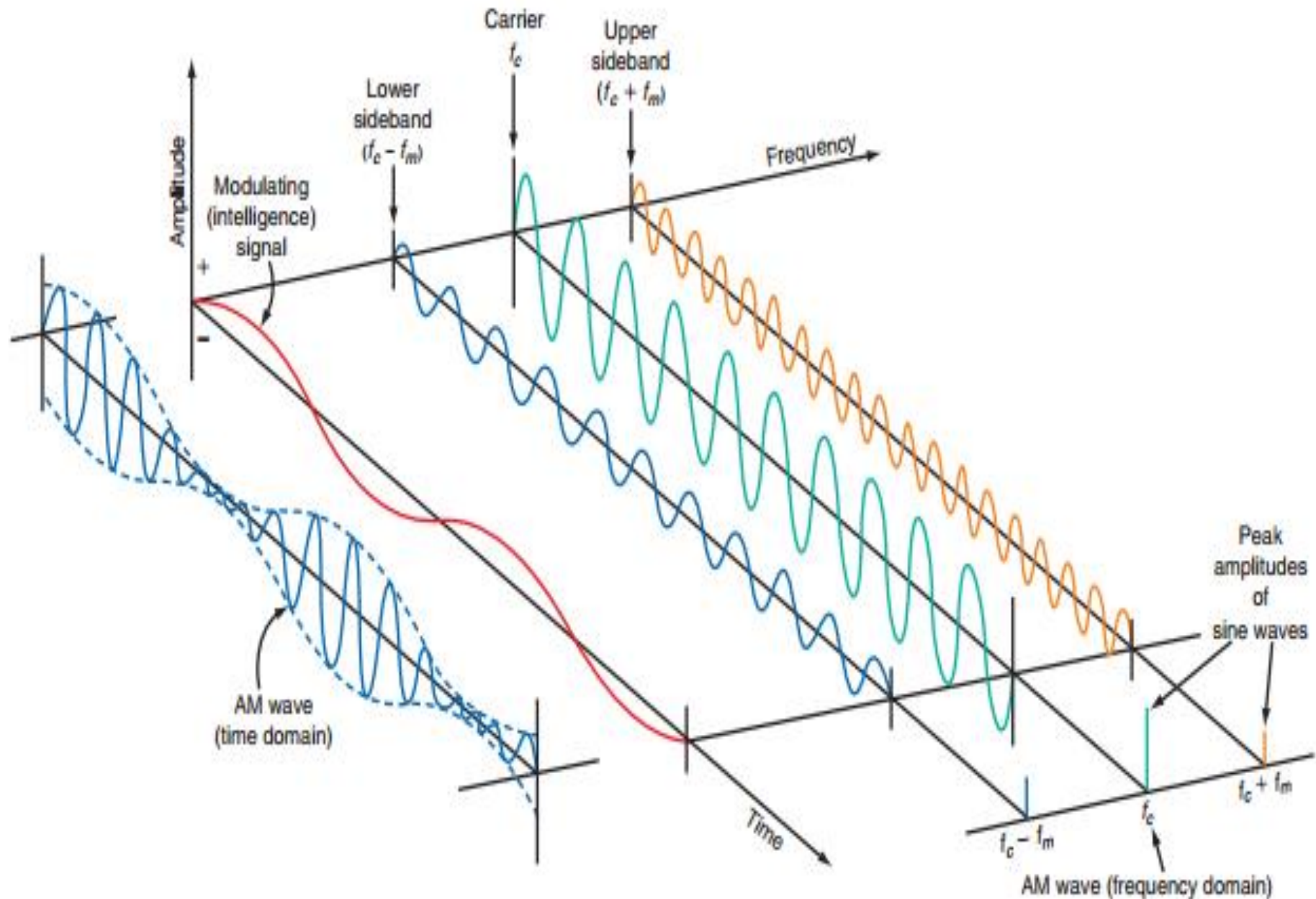
$$V_{\min} = V_c - V_m = 8 - 10 = \underline{\underline{-2 \text{ V}}}$$

$$m = \frac{20}{16} \times 100\% = 125\%$$

Frequency Spectrum of AM



Frequency Spectrum of AM



AM Derivation Practice

Q1. Assume $m(t) = A_m \sin \omega_m t$ and $c(t) = A_c \sin \omega_c t$

Derive expression for amplitude modulated (AM) wave AM(t).

Answer: $A_c \sin \omega_c t + \frac{A_c m_a}{2} \cos 2\pi(f_c - f_m)t - \frac{A_c m_a}{2} \cos 2\pi(f_c + f_m)t$

Q2. Assume $m(t) = A_m \sin \omega_m t$ and $c(t) = A_c \cos \omega_c t$

Derive expression for amplitude modulated (AM) wave AM(t).

Answer: $A_c \cos \omega_c t + \frac{A_c m_a}{2} \sin 2\pi(f_c - f_m)t - \frac{A_c m_a}{2} \sin 2\pi(f_c + f_m)t$

Q3. Assume $m(t) = A_m \cos \omega_m t$ and $c(t) = A_c \sin \omega_c t$

Derive expression for amplitude modulated (AM) wave AM(t).

Answer: $A_c \sin \omega_c t + \frac{A_c m_a}{2} \sin 2\pi(f_c + f_m)t + \frac{A_c m_a}{2} \sin 2\pi(f_c - f_m)t$